Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

Details of Invention for better understanding:

1. TITLE: Low-Cost Automatic Mechanical Flap-Based Wet and Dry Waste Segregation Dustbin for Household Use

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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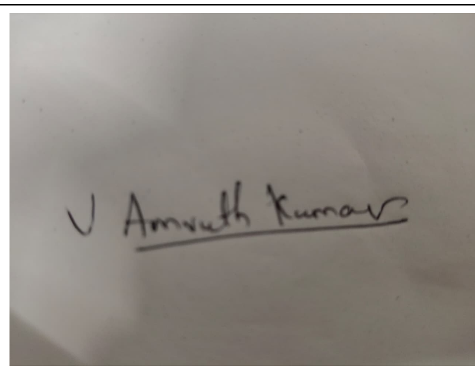
Signature (Mandatory)	
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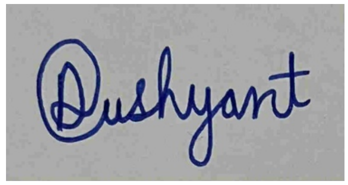
Signature (Mandatory)	
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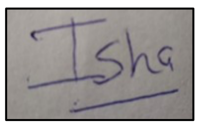
Signature (Mandatory)	

3. DESCRIPTION OF THE INVENTION:

- **1. Purpose**

The **Low-Cost Automatic Mechanical Waste Segregation Dustbin for Household Use** aims to simplify waste management at the household level by enabling automatic segregation of wet and dry waste without requiring user intervention, electricity, or complex electronic systems. This innovation addresses the limitations of traditional dustbins and smart dustbins by providing a cost-effective, maintenance-free, and user-friendly solution suitable for everyday domestic use.

2. Technical Workings

2.1 Single Entry Disposal System

The dustbin is designed with a single input slot where users can dispose of all types of household waste without manual segregation.

2.2 Mechanical Detection and Sorting Chamber

Once waste is dropped into the system, it passes through an internal mechanical sorting chamber that operates using passive physical principles:

- **Moisture-based detection mechanism:**
  Wet waste is identified using moisture-sensitive contact elements or absorbent-trigger pads.
- **Weight and gravity-based sorting:**
  Heavier or denser waste follows a different mechanical path compared to lighter waste.

2.3 Automated Flap-Based Diversion System

Based on the detected properties:

- Wet waste is directed into a **wet waste compartment** using an automatically triggered mechanical flap.
- Dry waste is routed into a **dry waste compartment** through an alternate channel.

The flap system operates using springs and gravity without electrical control.

2.4 Anti-Jamming and Self-Reset Mechanism

The system includes:

- Angled internal channels to prevent waste blockage
- Self-resetting spring flaps that return to default position after each operation
- Smooth internal surfaces to reduce clogging from wet waste

3. Unique Attributes

- **3.1 Fully Mechanical Automation**

Unlike existing smart dustbins that depend on sensors, AI, or electronics, this system operates entirely on **mechanical principles such as moisture response, weight distribution, and gravity-based motion**.

- **3.2 No User Decision Required**

Users do not need to segregate waste manually. The system automatically classifies waste at the point of entry, improving convenience and compliance.

- **3.3 Electricity-Free Operation**

The dustbin functions without any external power supply, making it suitable for rural areas, low-income households, and locations with limited infrastructure.

- **3.4 Low-Cost and Scalable Design**

The use of simple mechanical components such as flaps, springs, and angled channels ensures low manufacturing cost and easy scalability for mass household adoption.

- **3.5 Improved Waste Management Efficiency**

By separating wet and dry waste at the source, the system reduces contamination, improves recycling efficiency, and minimizes environmental impact.

- **4. Conclusion**

The proposed Low-Cost Automatic Mechanical Waste Segregation Dustbin provides an innovative and practical solution to household waste management challenges. By eliminating the need for electricity, sensors, or user decision-making, the system ensures reliable and automatic segregation of wet and dry waste. This invention enhances environmental sustainability, reduces waste contamination, and offers a highly scalable solution for widespread domestic use.

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A. PROBLEM ADDRESSED BY THE INVENTION:

The management of household waste presents several challenges that hinder efficient segregation and recycling of waste materials. The proposed automated mechanical flap-based dustbin addresses the following key problems:

1. **Improper Waste Segregation:**
   In most households, wet and dry waste are not separated at the source, leading to mixed waste that reduces recycling efficiency and increases landfill burden.
2. **Dependence on User Awareness:**
   Waste segregation relies heavily on user behavior, but lack of awareness and inconvenience often result in incorrect disposal practices.
3. **Lack of Low-Cost Automated Systems:**
   Existing smart dustbins are expensive and depend on electronic components, making them unsuitable for widespread household use.
4. **Maintenance and Complexity Issues:**
   Sensor-based or AI-based systems require regular maintenance and are prone to failure in real-world household conditions.
5. **Environmental Impact:**
   Improper segregation of waste leads to environmental pollution, poor recycling outcomes, and increased waste management challenges.
6. **Conclusion**
   The proposed invention addresses these issues by introducing a low-cost, fully mechanical system that automatically segregates wet and dry waste at the point of disposal without requiring user intervention or electrical power.

B. OBJECTIVE OF THE INVENTION (Provide minimum two)

- **Enable Automatic Waste Segregation:**

The primary objective of the invention is to develop a low-cost household dustbin that automatically separates wet and dry waste using a mechanical flap-based system, eliminating the need for manual segregation by users.

- **Reduce Dependence on User Behavior:**

Another key objective is to minimize reliance on user awareness and decision-making during waste disposal by providing an automated system that correctly classifies waste at the point of entry, ensuring consistent and accurate segregation.

- **Promote Low-Cost and Sustainable Solution:**

The invention aims to provide an affordable and electricity-free waste management solution suitable for all households, including rural and low-income areas, thereby encouraging widespread adoption and improving environmental sustainability.

- **Improve Waste Management Efficiency:**

A further objective is to enhance recycling efficiency and reduce environmental pollution by ensuring proper separation of waste at the source, thereby reducing contamination and improving downstream waste processing.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY:

Sr. No	Patent I'd	Abstract	Research Gap	Novelty
1.	US Patent – Smart Waste Bin Systems	This patent describes smart waste bins that use sensors and microcontrollers to detect waste type and open corresponding compartments automatically.	These systems depend on electronics, require power supply, and are costly and unsuitable for large-scale household deployment.	The proposed invention introduces a **fully mechanical, electricity-free flap-based system** for automatic wet and dry waste segregation, reducing cost and maintenance.
2.	EP Patent – Automated Waste Sorting System	This invention focuses on automated waste sorting using AI vision systems and robotic mechanisms for classification of waste materials.	AI and robotic systems are expensive, complex, and require high maintenance, making them impractical for daily household use.	The proposed system eliminates AI dependency and uses **passive mechanical principles like gravity, moisture response, and weight-based separation** for sorting waste.
3.	Indian Patent – Sensor-Based	This patent describes a dustbin that uses sensors to detect waste level	The system is dependent on sensors and electronics, which increases cost and reduces reliability in real-	The invention provides a **low-cost, robust mechanical flap**

	Dustbin for Waste Management	and categorize waste into different bins using electronic control systems.	world dusty and wet environments	**system that operates without sensors or electronics, ensuring reliability and easy maintenance in households.**

Conclusion

The proposed invention fills critical gaps in existing patented systems by eliminating the need for sensors, AI, and electrical components. It introduces a low-cost, fully mechanical waste segregation system that is reliable, maintenance-free, and suitable for widespread household use, ensuring efficient wet and dry waste separation at the source.

D. DETAILED DESCRIPTION:

The **Low-Cost Automatic Mechanical Waste Segregation Dustbin** is an innovative solution designed to improve household waste management by enabling automatic segregation of wet and dry waste using a fully mechanical system. The invention eliminates the need for electricity, sensors, or user intervention, ensuring a simple, reliable, and cost-effective operation suitable for everyday domestic use.

2. System Components

2.1 Outer Dustbin Structure

- **Main Body:**
  A durable container made of plastic or lightweight material, designed to hold multiple internal compartments.
- **Single Entry Opening:**
  A top input slot where users can dispose of waste without manual segregation.

  .
- **2.2 Mechanical Sorting Chamber**

- **Detection Zone:**
  A small internal chamber where waste initially lands and its physical properties (such as moisture and weight) are passively assessed.

- **Moisture Contact Surface:**
  A surface or pad that comes into contact with waste to help differentiate wet waste from dry waste based on moisture presence.

2.3 Flap-Based Diversion System

- **Mechanical Flaps:**
  Movable flaps positioned inside the bin that divert waste into appropriate compartments.

- **Spring Mechanism:**
  Ensures that the flap returns to its default position after each operation.

- **Gravity Channels:**
  Angled internal pathways that guide waste smoothly into designated compartments.

2.4 Waste Collection Compartments

- **Wet Waste Compartment:**
  Collects biodegradable and moist waste such as food scraps.

- **Dry Waste Compartment:**
  Stores non-biodegradable materials such as paper, plastic, and packaging.

3. Technical Functionality

- **3.1 Automatic Waste Segregation Process**

- **Waste Entry:**
  The user disposes of waste through a single opening without any prior sorting.

- **Initial Contact:**
  Waste enters the detection chamber and comes into contact with the moisture-sensitive surface.

- **Decision Mechanism:**
  - If moisture is detected → flap redirects waste to the wet compartment
  - If no moisture is detected → waste follows default path to dry compartment

3.2 Mechanical Flap Operation

- The flap is triggered based on physical interaction (moisture/weight/gravity).
- It opens momentarily to direct waste into the correct compartment.
- A spring mechanism resets the flap automatically for the next use.

3.3 Anti-Jamming Mechanism

- Smooth and angled internal surfaces prevent waste from sticking.
- Gravity-assisted movement ensures continuous flow without blockage.
- Minimal moving parts reduce chances of mechanical failure.

4. Unique Features

4.1 Electricity-Free Operation

- The system operates entirely on mechanical principles such as gravity and spring action, eliminating the need for power supply.

4.2 Automatic Segregation Without User Intervention

- Users are not required to manually separate waste, improving ease of use and accuracy.

4.3 Low-Cost and Scalable Design

- The use of simple materials and components makes the system affordable and suitable for large-scale household adoption.

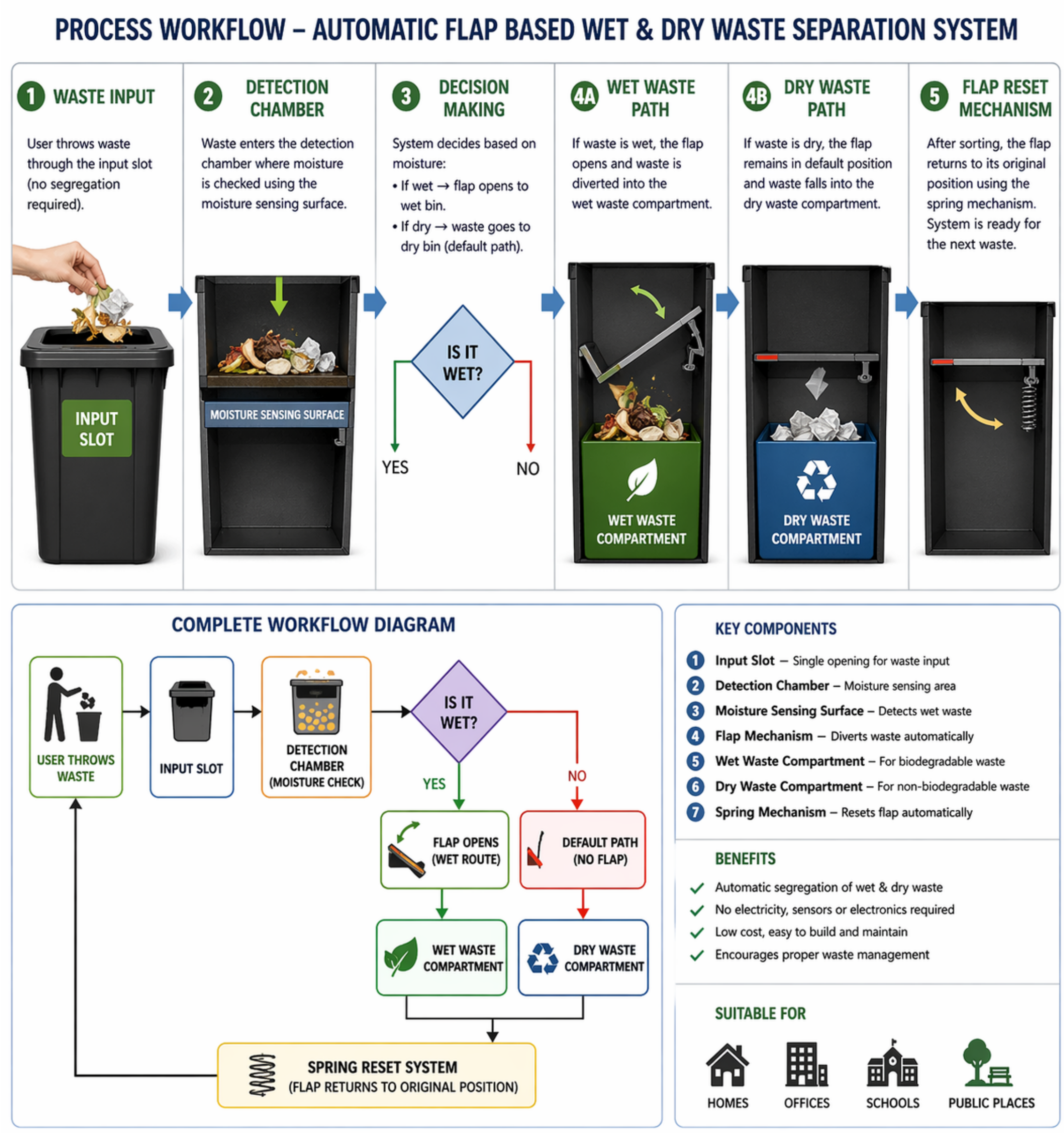
4.4 Low Maintenance Requirement

- Absence of sensors and electronics reduces maintenance needs and increases reliability in real-world conditions.

- **Conclusion**

The Low-Cost Automatic Mechanical Waste Segregation Dustbin offers a practical and efficient solution to household waste management challenges. By integrating a simple flap-based mechanical system, it enables automatic segregation of wet and dry waste without electricity or user effort. This innovation improves recycling efficiency, reduces environmental impact, and provides a scalable solution for widespread domestic use.

Process Workflow:



- **E. RESULTS AND ADVANTAGES**

The **Low-Cost Automatic Mechanical Waste Segregation Dustbin** provides several important results and advantages that make it superior to traditional dustbins and existing smart waste systems.

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1. Improved Waste Segregation Efficiency

- **Automatic Segregation:**
  The system separates wet and dry waste at the point of disposal without requiring user intervention, improving accuracy and consistency.
- **Reduced Waste Contamination:**
  Proper separation prevents mixing of waste, increasing recycling efficiency and improving composting outcomes.

2. Low-Cost and Affordable Solution

- **Cost-Effective Design:**
  The use of simple mechanical components such as flaps, springs, and gravity-based channels significantly reduces manufacturing costs.
- **Suitable for All Households:**
  The affordability makes the system accessible for rural, urban, and low-income households.

3. Electricity-Free Operation

- **No Power Requirement:**
  The system operates entirely on mechanical principles, eliminating dependency on electricity.
- **Energy Efficient:**
  No energy consumption reduces operational costs and supports sustainable living.

4. Reduced Maintenance

- **Minimal Mechanical Components:**
  The absence of sensors, electronics, and motors reduces chances of failure.
- **Long-Term Reliability:**
  Simple design ensures durability and consistent performance in real-world conditions.

- **5. User Convenience**
- **No User Effort Required:**
  Users can dispose of waste normally without worrying about segregation.
- **Time-Saving:**
  Eliminates the need for manual sorting, making waste disposal quick and easy.

- **6. Environmental Benefits**
- **Improved Recycling Efficiency:**
  Proper segregation at source enhances downstream recycling processes.
- **Reduced Pollution:**
  Prevents contamination of recyclable waste and reduces landfill burden.

- **7. Enhanced Safety and Hygiene**
- **Better Waste Handling:**
  Separating wet waste reduces odor and bacterial growth.
- **Cleaner Environment:**
  Organized waste storage improves hygiene in households.

Comparison to Existing Prior Art

- **Traditional Dustbins:**
  Do not provide any segregation, leading to mixed waste and poor recycling outcomes.
  → The proposed system enables **automatic segregation**.
- **Sensor-Based Smart Dustbins:**
  Expensive, require electricity, and need regular maintenance.
  → The proposed system is **low-cost, mechanical, and maintenance-free**.

- **Manual Segregation Systems:**
  Depend on user awareness and effort.
  → The proposed system removes **user dependency completely**.

- **Conclusion**

The Low-Cost Automatic Mechanical Waste Segregation Dustbin offers a practical, efficient, and affordable solution to household waste management. By combining automatic segregation, low cost, and electricity-free operation, the system improves waste handling efficiency while promoting environmental sustainability. It provides a scalable and reliable alternative to both traditional and existing smart dustbin systems.

E. EXPANSION:

To ensure effective implementation and wide adoption of the Low-Cost Automatic Mechanical Waste Segregation Dustbin, several key variables must be considered. These factors influence the design, performance, and scalability of the system.

- **1. Waste Type Compatibility**
- **Variety of Household Waste:**
  The system should handle different types of waste such as food waste, paper, plastic, and mixed materials without failure.
- **Moisture Variation:**
  The detection mechanism should work effectively for different moisture levels in wet waste.

- **2. Mechanical Design Efficiency**
- **Flap Sensitivity and Accuracy:**
  The flap mechanism must respond correctly to waste properties to ensure proper segregation.
- **Durability of Components:**
  Materials used for flaps, springs, and internal structures must withstand continuous usage.

- **3. Structural Design and Size**
- **Compact Design:**
  The dustbin should be suitable for household spaces such as kitchens and small rooms.
- **Capacity Variations:**
  Different sizes may be designed for homes, offices, or public spaces.

- **4. Material Selection**
- **Cost-Effective Materials:**
  Use of low-cost plastics or recycled materials to maintain affordability.
- **Corrosion and Moisture Resistance:**
  Materials should resist damage from wet waste and environmental conditions.

- **5. User Interaction and Convenience**
- **Ease of Use:**
  The system must allow users to dispose of waste without additional effort or instructions.
- **Ease of Cleaning:**
  Internal compartments should be easy to remove and clean.

- **6. Maintenance and Reliability**
- **Low Maintenance Requirement:**
  The system should function with minimal servicing due to absence of electronics.
- **Clog Prevention Mechanism:**
  Design should prevent blockage caused by sticky or bulky waste.

- **7. Environmental Impact**
- **Support for Recycling:**
  Proper segregation improves recycling and composting processes.
- **Sustainability:**
  Use of recyclable materials and reduction in energy usage supports eco-friendly goals.

- **8. Scalability and Adaptability**
- **Mass Production Feasibility:**
  The design should be simple enough for large-scale manufacturing.
- **Adaptability:**
  The system can be modified for use in households, schools, offices, and public places.

- **Conclusion**

By addressing these variables, the proposed mechanical waste segregation dustbin can achieve reliable performance, low cost, and wide usability. Its scalable and adaptable design makes it suitable for large-scale implementation, contributing to improved waste management and environmental sustainability.

F. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

Working prototype is not ready. It will take at least a year to complete it.

- **G. EXISTING DATA**

To support the **Low-Cost Automatic Mechanical Waste Segregation Dustbin**, it is important to consider existing data and studies related to waste management, segregation practices, and environmental impact. The following data strengthens the need and effectiveness of the proposed invention:

- **1. Waste Segregation Practices**
- **Low Segregation at Source:**
  Studies in urban and rural areas show that a large percentage of households do not segregate waste at the source due to lack of awareness and convenience.
- **Manual Segregation Issues:**
  Data indicates that reliance on manual segregation leads to inconsistent results and high levels of mixed waste.

- **2. Environmental Impact of Improper Waste Disposal**
- **Increased Landfill Waste:**
  Mixed waste contributes significantly to landfill overflow, reducing the efficiency of recycling systems.
- **Pollution and Health Risks:**
  Improper waste segregation leads to soil, water, and air pollution, as well as increased health risks due to bacterial growth and toxic exposure.

- **3. Limitations of Existing Smart Dustbins**
- **High Cost of Sensor-Based Systems:**
  Existing smart bins using sensors and AI are expensive and not suitable for mass household adoption.
- **Maintenance Challenges:**
  Data shows that electronic systems often fail in dusty and wet conditions, requiring frequent maintenance.

- **4. Benefits of Source-Level Segregation**
- **Improved Recycling Efficiency:**
  Proper segregation at the source can significantly increase recycling rates and reduce processing costs.
- **Better Composting Outcomes:**
  Separating wet waste improves compost quality and efficiency in waste treatment processes.

- **5. User Behavior and Convenience**
- **Lack of User Participation:**
  Surveys indicate that users are less likely to segregate waste if additional effort is required.
- **Need for Automated Solutions:**
  Systems that reduce user effort have higher adoption rates and better compliance.

- **6. Comparative Analysis of Waste Management Systems**
- **Traditional Dustbins:**
  Do not provide segregation, leading to inefficient waste management.
- **Smart Dustbins:**
  Provide automation but are costly, complex, and less reliable in household conditions.
- **Mechanical Systems (Proposed):**
  Offer a balance of automation, low cost, and reliability without requiring electricity or maintenance.

- **Conclusion**

Existing data clearly highlights the need for a simple, reliable, and low-cost waste segregation system at the household level. The proposed mechanical dustbin addresses the limitations of both traditional and smart systems by providing automatic segregation without user effort or electrical dependency. This makes it a practical and scalable solution for improving waste management and environmental sustainability.

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?		NO (No)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?		NO (No)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?		NO (No)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.		NO (No)
E. Name of Regulatory body or any other approvals if required.		NO (No)

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. **NA**

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity). **NA**

7. Potential Chances of Commercialization. **Yes**

8. List of companies which can be contacted for commercialization along with the website link.

Here are two companies that specialize in electric vehicle charging solutions and could be potential partners for the commercialization of the Self-Sufficient Charging System for Electric Public Transportation:

- **1. Nilkamal Limited**
- **Overview:** One of India's largest manufacturers of plastic products, including household dustbins and storage solutions. They have strong distribution networks across India.
- **Website:** https://www.nilkamal.com

- **2. Supreme Industries Limited**
- **Overview:** A leading plastic manufacturing company producing a wide range of household and industrial plastic products, including waste management solutions.
- **Website:** https://www.supreme.co.in

9. Any basic patent which has been used and we need to pay royalty to them.

10**. FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings - (Provisional)

11. **KEYWORDS:**

• Waste Segregation
• Mechanical Dustbin System
• Automated Flap Mechanism
• Wet and Dry Waste Separation
• Household Waste Management
• Low-Cost Dustbin

- Electricity-Free System
- Passive Mechanical Design
- Source-Level Segregation
- Sustainable Waste Management
- Recycling Efficiency
- Environmental Protection
- User-Friendly Waste Disposal
- Gravity-Based Sorting
- Spring Mechanism
- Smart Dustbin (Mechanical)
- Eco-Friendly Solution
- Waste Reduction
- Hygienic Waste Handling
- Scalable Waste Solution

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

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